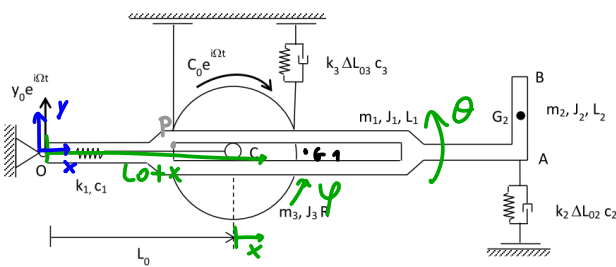


12/12/15-B

D.o.f.



$x = \begin{bmatrix} x \\ \theta \\ y \end{bmatrix}$ $x_0 = \begin{bmatrix} L_0 \\ 0 \\ 0 \end{bmatrix}$

$2 \text{ R.B} \rightarrow 6 \text{ dof}$
 $\text{wing.co} \rightarrow -2 \text{ dof}$
 $1 \text{ string} \rightarrow -1 \text{ dof}$
 $1 \text{ cart} \rightarrow -1 \text{ dof}$

$2 \text{ dof} + y$
 $\downarrow$
 x, y

$$x=0, \theta \neq 0 \rightarrow p = \frac{L\theta}{R} \quad (x_c = L\theta \text{ and } p = U\theta)$$

$x \neq 0, \theta = 0 \rightarrow \psi = 0$ (translatory motion)

$$X_C = (L_0 + x) \cos \theta \approx (L_0 + x) \left(1 - \frac{\theta^2}{2}\right) = \cancel{L_0} + x - L_0 \frac{\theta^2}{2}$$

$$y_L = (l_0 + x) \sin \theta + y \approx l_0 \theta + y + x \theta$$

$$x_{G_1} = \frac{l_1}{2} \cos \theta \approx \frac{l_1}{2} - \frac{l_1}{2} \frac{\theta^2}{2}$$

$$y_{G1} = \frac{L_1}{2} \sin \theta + y \approx \frac{L_1}{2} \theta + y$$

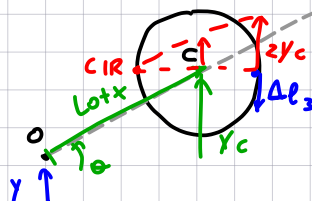
$$X_{G2} = L_1 \cos \theta - \frac{L_2}{2} \sin \theta \approx L_1 - \frac{L_2}{2} \theta - \frac{L_2}{2} \frac{\theta^2}{2}$$

$$y_{t2} = L_1 \sin \theta + \frac{L_2}{2} \cos \theta + y \approx \frac{L_2}{2} + L_1 \theta + y - \frac{L_2}{2} \frac{\theta^2}{2}$$

$$\Delta l_1 = x$$

$$\Delta l_2 = y_A = l_1 \sin \theta + y \approx l_1 \theta + y$$

$$\Delta l_3 = -2y_c \approx -2l_0 \theta - 2y - 2x\theta$$



$$T = \frac{1}{2} m_1 \dot{x}_{01}^2 + \frac{1}{2} m_1 \dot{y}_{01}^2 + \frac{1}{2} J_1 \dot{\theta}^2 + \frac{1}{2} m_2 \dot{x}_{02}^2 + \frac{1}{2} m_2 \dot{y}_{02}^2 + \frac{1}{2} J_2 \dot{\theta}^2 + \frac{1}{2} m_3 \dot{x}_c^2 + \frac{1}{2} m_3 \dot{y}_c^2 + \frac{1}{2} J_3 \dot{\varphi}^2$$

$$V = \frac{1}{2} \sum_{i=1}^3 k_i \cdot 4 \ell_i^2 + \sum_{i=1}^3 m_i g h_i$$

$$D = \frac{1}{2} \sum_{i=1}^3 c_i \Delta x_i^2$$

$$\delta W_n = -\omega_c^2 \int y \, dy + F_y \delta y$$

Jacobian matrices

$$[A_w] = \begin{bmatrix} x & \theta & y \\ 0 & 0 & 0 \\ 0 & \frac{L_1}{2} & 1 \\ 0 & 1 & 0 \\ 0 & -\frac{L_2}{2} & 0 \\ 0 & L_1 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & L_0 & 1 \\ 0 & \frac{L_0}{R} & 0 \end{bmatrix} \begin{matrix} x_{w1} \\ y_{w1} \\ \theta \\ x_{w2} \\ y_{w2} \\ \theta \\ x_c \\ y_c \\ y \end{matrix}$$

$$[A_n] = \begin{bmatrix} x & \theta & y \\ 1 & 0 & 0 \\ 0 & L_1 & 1 \\ 0 & -2L_0 & -2 \end{bmatrix} \begin{matrix} \Delta l_1 \\ \Delta l_2 \\ \Delta l_3 \end{matrix}$$

$$[L_q] = \begin{bmatrix} x & \vartheta & y \\ 0 & \frac{L_0}{R} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{matrix} dx \\ d\varphi \\ dy \end{matrix}$$

1/R (phi depends on y)

Stiffness matrices

$$k_{g2} = m_{g2} \begin{bmatrix} 0 & 0 & 0 \\ 0 & -\frac{k_z}{2} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$k_3 = m_3 g \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$K_{1,2} = K_{3,4} \Delta l_{03} \begin{bmatrix} 0 & -2 & 0 \\ -2 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

② MATLAB

③ OUTPUT

$$\varphi = \frac{L_0}{R} \theta; \quad y_c = l\theta + y$$

$$Q_F = \Lambda_q(1; 1:2)' \cdot (-1) \quad \uparrow \quad \omega = 1$$

$$\Lambda_q^T = \begin{bmatrix} 0 & 0 \\ \frac{L_0}{R} & 0 \\ 0 & 1 \end{bmatrix}$$

$$Q_F = \begin{bmatrix} 0 & 0 \\ \frac{L_0}{R} & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} -C_0 e^{i\omega t} \\ F_y \end{bmatrix} = \begin{bmatrix} -C_0 e^{i\omega t} \\ -\frac{L_0}{R} C_0 e^{i\omega t} \\ F_y \end{bmatrix}$$

Q_F
 Q_c

④ OUTPUT

$$x_c = x; \quad F_{cl_3} = (k_3 + i\omega l_3) \Delta l_3$$

$$(\Delta l_3 = -2l_0\theta - 2y_0) \quad \uparrow \quad y_0 = 1$$

$$Q_F = 0$$

$$Q_{Fc} = -(-\omega^2 [m_{cf}] + i\omega [c_{cf}] + [k_{cf}]) \cdot 1$$

$$\rightarrow x = A^{-1} Q_{Fc}$$

from (

$$[m_{ff}] \ddot{x}_f + [c_{ff}] \dot{x}_f + [k_{ff}] x_f = Q_F - ([m_{fc}] \ddot{x}_c + [c_{fc}] \dot{x}_c + [k_{fc}] x_c)$$

A Q_{Fc} $x_c = y_0 e^{i\omega t}$

⑤ OUTPUT

$$V_0 = F_y = Q_c$$

1 $C_0 \neq 0, y_0 = 0$ both y_{02} and y_{03}

$$[m_{ff}] \ddot{x}_f + [c_{ff}] \dot{x}_f + [k_{ff}] x_f = Q_F - ([m_{fc}] \ddot{x}_c + [c_{fc}] \dot{x}_c + [k_{fc}] x_c)$$

A Q_{Fc} $x_c = y_0 e^{i\omega t}$

$$x = A^{-1} Q_F \quad \leftarrow \quad Q_F = \Lambda_q(1; 1:2)' (-C_0 e^{i\omega t})$$

$$V_0 = Q_c = [m_{cf}] \ddot{x}_f + [c_{cf}] \dot{x}_f + [k_{cf}] x_f + [m_{cc}] \ddot{x}_c + [c_{cc}] \dot{x}_c + [k_{cc}] x_c$$

FRF₁: $V_0 = (-\omega^2 [m_{cf}] + i\omega [c_{cf}] + [k_{cf}]) \cdot x$

2 $C_0 = 0, y_{02} \neq 0, y_{03} = 0$

$$[m_{ff}] \ddot{x}_f + [c_{ff}] \dot{x}_f + [k_{ff}] x_f = Q_F - ([m_{fc}] \ddot{x}_c + [c_{fc}] \dot{x}_c + [k_{fc}] x_c)$$

A Q_{Fc} $x_c = y_{02} e^{i\omega t}$

$$x = A^{-1} Q_{Fc}$$

$$Q_c = V_0 = [m_{cf}] \ddot{x}_f + [c_{cf}] \dot{x}_f + [k_{cf}] x_f + [m_{cc}] \ddot{x}_c + [c_{cc}] \dot{x}_c + [k_{cc}] x_c$$

FRF₂: $V_0 = (-\omega^2 [m_{cf}] + i\omega [c_{cf}] + [k_{cf}]) \cdot x + (-\omega^2 [m_{cc}] + i\omega [c_{cc}] + [k_{cc}]) x_c$

3 $C_0 = 0, y_{02} = 0, y_{03} \neq 0$

$$[m_{ff}] \ddot{x}_f + [c_{ff}] \dot{x}_f + [k_{ff}] x_f = Q_F - ([m_{fc}] \ddot{x}_c + [c_{fc}] \dot{x}_c + [k_{fc}] x_c)$$

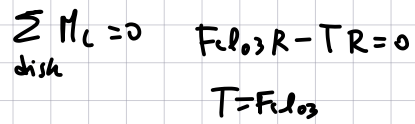
A Q_{Fc} $x_c = y_{03} e^{i\omega t}$

$$x = A^{-1} Q_{Fc}$$

$$Q_c = V_0 = [m_{cf}] \ddot{x}_f + [c_{cf}] \dot{x}_f + [k_{cf}] x_f + [m_{cc}] \ddot{x}_c + [c_{cc}] \dot{x}_c + [k_{cc}] x_c$$

FRF₃: $V_0 = (-\omega^2 [m_{cf}] + i\omega [c_{cf}] + [k_{cf}]) \cdot x + (-\omega^2 [m_{cc}] + i\omega [c_{cc}] + [k_{cc}]) x_c$

$$\Rightarrow FRF_{TOT} = FRF_1 + FRF_2 + FRF_3 \quad (2 \text{ period } T_{MAX} = \frac{2\pi}{\omega_3} \rightarrow t = [0; 0.001; \frac{2T_{MAX}}{1.8}])$$



$$F_{\text{el},3}(L_0 + \cancel{r}) + T(L_0 - \cancel{r}) - L_0 m_3 g - b m_2 g - L_1 F_{\text{el},2}$$

$$\Rightarrow F_{cl03} = \frac{m3g}{2} + \frac{L1 F_{cl02} + b m2g}{2L0}$$

$$b = \sqrt{L_1^2 + L_2^2}$$